

1. System Architecture Overview

The application operates as a Single Page Application (SPA), allowing the frontend to interact seamlessly and directly with the Firebase backend.

- **Frontend:** Standard Web Technologies (HTML, CSS, JS) combined with Bootstrap for responsive UI design.
- **Backend:** Firebase Firestore (NoSQL Database) for efficient storage of users, products, sales, and operational data.
- **Authentication:** Firebase Auth handles secure login for both staff and customers.
- **Access Levels:**
 - **Admin/Employees:** Access the Management Dashboard (POS, Inventory, Reports).
 - **Customers:** Access the Customer Portal (Browse products, View orders).

2. Module Breakdown & Facilities

A. Point of Sale (POS) & Sales Management

File Name	Core Facilities
pos.html	Core cashier interface. Real-time product search/scanning, cart management, customer selection, multi-mode pricing (Retail/Wholesale), discounts, wallet/point redemption, and receipt printing.
pos-sales-history.html	View past transactions, reprint receipts, edit recent sales, and process returns.
order-sale-history.html	Manage online orders from the portal. Update status (Pending → Delivered).
sell-return.html & sell-return-history.html	Process customer returns, issue refunds to cash/wallet, and automatically restock items.
due-history.html	Track unpaid invoices (credit sales) and process due payments.
wallet-sale-history.html	Specific transaction log for purchases made via the customer's digital wallet.

B. Inventory & Product Management

File Name	Core Facilities
web.html	Master product list. View stock levels, check prices, and edit product details.
stock-report.html	Detailed view of inventory value and quantity. Supports full stock report printing.
stock-adjust.html	Manual stock correction (e.g., for damaged goods or counts) without a sale/purchase.
purchase.html & purchase-history.html	Record supplier intake, update inventory quantities, and track cost prices.
purchase-return.html & purchase-return-history.html	Return defective goods to suppliers and adjust stock records accordingly.

C. Customer Relationship Management (CRM)

File Name	Core Facilities
customer.html	Customer database. Track contact info, loyalty points, wallet balance, and total spend.
print-customar-card.html	Generate printable membership cards with barcodes/QR codes.
supplier.html	Manage supplier details and track purchase history per supplier.

D. Financials & Reporting

File Name	Core Facilities
dashboard.html	High-level analytics: sales trends, top products, low stock alerts, and financial summaries.
day-report.html & day-report-history.html	End-of-day closing: calculates opening cash, total sales, expenses, and final drawer cash.
monthly-expenses.html & monthly-expenses-history.html	Track operational costs (Rent, Salaries, Electricity) to assist in net profit calculation.
profit-and-loss.html	Calculates Gross Profit (Sales Revenue - Cost of Goods Sold).

File Name	Core Facilities
leaderboard.html	Gamification dashboard showing top employees, best-selling products, and top customers.

E. Administration & Security

File Name	Core Facilities
employee.html	User management. Create staff accounts and assign granular permissions.
signin.html	Secure login portal with rate limiting to prevent brute-force attacks.
restricted.html & restriction.js	Security protocols to prevent unauthorized code inspection or window resizing (Kiosk mode).

F. Customer Portal (Client Side)

File Name	Core Facilities
customar-portal.html	Customer dashboard displaying points, wallet balance, and recent activity.
customar-products.html & customar-product-details.html	E-commerce interface for browsing items and cart management.
customar-orders.html	Order status tracking for customers.
customar-recovery.html	Password recovery via phone number OTP verification.

3. Operational Workflows

Inventory Flow

1. **Stock In:** purchase.html → Item added → Products collection updated (Stock +).
2. **Stock Out:** pos.html OR customar-orders.html → Sale processed → Products collection updated (Stock -).
3. **Stock Adjustment:** stock-adjust.html → Manual stock correction applied.

Sales Flow

- **In-Shop (Staff):** Log in → Open pos.html → Scan item → Select customer → Process payment (Cash/Wallet) → Print receipt & Update Database.

- **Online (Customer):** Log in → Open customar-products.html → Add to cart → Checkout → Order routed to order-sale-history.html for staff fulfillment.

Financial Flow

1. **Daily Accumulation:** Sales automatically accrue in the sales collection.
 2. **Daily Closing:** Staff opens day-report.html → System runs calculations (*Opening Cash + Sales - Expenses*) → Report is saved.
 3. **Monthly Reconciliation:** monthly-expenses.html logs overhead (rent/salary) → profit-and-loss.html calculates the final Net Profit.
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4. Key Features Summary

- **Granular Permissions:** Precise control over employee access (e.g., cashiers can process sales but cannot view profit margins).
- **Dual-Interface Design:** Distinct, optimized environments for shop staff (Admin/POS) and end-users (Customer Portal).
- **Digital Wallet & Loyalty Program:** Integrated system for managing store credit, refunds, and reward points.
- **Offline Resilience:** Utilization of Service Workers (firebase-init.js) to provide Progressive Web App (PWA) offline capabilities.
- **Dynamic Pricing Engine:** Automatic price tier adjustments (Retail vs. Wholesale) triggered by the assigned customer type.

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